

CASPER NATRONA COUNTY INTL, WY, USA

WMO: 725690

Lat: 42.898N

Lon: 106.474W

Elev: 1621

StdP: 83.31

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24089

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-22.2	-18.3	-25.4	0.5	-21.8	-22.0	0.6	-17.5	17.5	4.2	15.7	2.7	3.1	30	0.722

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	34.5	15.2	33.0	14.9	31.5	14.6	17.4	28.0	16.6	27.6	15.9	27.2	5.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.3	12.4	19.2	13.1	11.5	18.9	12.1	10.7	18.7	54.7	27.7	52.3	27.4	50.0	26.9	20.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.5	12.6	11.4	-28.6	37.0	3.8	1.0	-31.3	37.7	-33.5	38.3	-35.7	38.9	-38.4	39.6		
(5)			-28.8	18.8	3.7	0.8	-31.5	19.4	-33.7	19.9	-35.7	20.4	-38.4	21.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.9	-3.3	-2.7	2.3	5.9	11.1	17.1	22.0	20.8	15.3	7.6	1.6	-3.7		
	DBStd	10.44	6.26	6.58	6.01	5.05	4.63	4.00	2.85	3.11	4.79	5.54	6.56	6.79		
	HDD10.0	2004	414	357	244	140	42	3	0	0	12	109	257	426		
	HDD18.3	4076	672	589	498	374	226	67	6	13	112	332	502	684		
	CDD10.0	1232	0	0	5	16	77	217	372	334	170	36	5	0		
	CDD18.3	262	0	0	0	0	2	31	119	89	20	1	0	0		
	CDH23.3	4685	0	0	1	9	122	752	1868	1441	468	24	0	0		
	CDH26.7	2092	0	0	0	0	24	317	937	654	158	2	0	0		
(14) Wind	WSAvg	5.2	6.7	6.1	5.6	5.1	4.6	4.4	4.0	4.1	4.4	5.0	6.0	6.7		
(15) Precipitation	PrecAvg	312	13	15	22	37	54	36	32	19	24	27	18	15		
	PrecMax	481	36	36	58	100	164	119	90	68	86	117	69	35		
	PrecMin	167	1	4	3	2	0	2	2	1	2	0	1	2		
	PrecStd	72	8	7	12	21	37	29	20	15	21	22	12	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	13.7	20.9	24.7	29.4	35.1	36.7	35.3	32.6	26.3	19.7	12.2		
		MCWB	4.3	5.8	7.9	10.5	13.0	14.4	16.0	15.4	14.9	11.2	7.5	4.6		
	2%	DB	8.4	10.8	17.6	21.7	26.6	32.8	35.0	33.6	30.5	23.2	16.2	9.4		
		MCWB	2.5	4.0	6.7	8.9	12.3	14.4	15.6	14.8	13.5	10.1	6.3	3.3		
	5%	DB	6.5	8.4	14.9	19.1	24.1	30.6	33.6	32.2	28.5	20.5	13.4	7.2		
		MCWB	1.6	2.8	5.6	7.9	11.7	14.0	15.4	14.7	12.7	9.1	5.2	2.0		
	10%	DB	4.8	6.1	12.0	16.3	21.6	28.0	32.0	30.6	26.1	17.6	10.8	5.3		
		MCWB	0.6	1.4	4.4	6.9	10.9	13.6	15.0	14.5	12.2	8.1	3.9	0.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.8	6.2	8.6	11.3	15.0	17.5	18.7	17.9	16.6	12.8	8.1	5.0		
		MCDB	9.8	13.0	19.0	22.7	25.7	27.2	30.2	28.6	28.3	22.3	17.4	11.2		
	2%	WB	3.1	4.4	7.2	9.6	13.5	16.2	17.7	16.8	15.1	10.9	6.7	3.6		
		MCDB	7.4	10.2	16.2	19.6	23.3	26.9	28.4	28.2	26.2	20.7	15.3	8.7		
	5%	WB	1.9	3.0	5.9	8.4	12.5	15.3	16.9	16.1	13.9	9.7	5.5	2.3		
		MCDB	6.1	7.9	14.0	17.9	22.1	26.3	27.7	27.5	24.9	19.4	12.6	6.9		
	10%	WB	0.7	1.6	4.8	7.3	11.4	14.5	16.3	15.4	12.9	8.5	4.3	1.0		
		MCDB	4.6	5.9	11.7	15.5	20.4	25.1	27.4	27.0	23.4	17.0	10.5	4.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	11.3	13.4	14.2	15.2	17.8	18.7	18.2	17.1	14.5	12.1	10.2		
		MCDBR	10.8	12.7	17.0	19.6	19.9	22.0	21.1	20.3	20.6	18.9	15.2	11.1		
	5% WB	MCWBR	6.8	7.7	8.5	9.4	8.3	7.9	6.7	6.8	7.8	9.0	8.1	6.9		
		MCWBR	10.5	12.5	16.3	18.0	18.2	19.2	18.2	17.7	17.6	17.8	14.5	11.0		
(40) Clear-Sky Solar Irradiance	taub		0.237	0.247	0.267	0.299	0.317	0.319	0.346	0.341	0.303	0.270	0.250	0.239		
	taud		2.553	2.543	2.527	2.454	2.460	2.469	2.420	2.418	2.521	2.591	2.559	2.540		
	Ebn at Noon		939	979	989	971	955	950	919	914	934	936	911	902		
	Edh at Noon		65	78	91	106	109	108	113	109	91	74	64	60		
(44) All-Sky Solar Radiation	RadAvg		2.06	2.90	4.18	5.25	6.22	7.44	7.31	6.34	5.08	3.40	2.23	1.64		
	RadStd		0.12	0.18	0.34	0.31	0.51	0.47	0.26	0.17	0.35	0.26	0.18	0.09		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page